

a) Projiziona Gammern. Schmidt'sche orthogonalisation in $\text{lin}(x) = (1, x, x^2)$

$$\vec{u}_1: \|\vec{u}_1\| = \sqrt{\int_1^5 1 \, dx} = \sqrt{4} = 2$$

$$\vec{u}_1 = \frac{1}{\|\vec{u}_1\|} = \frac{1}{2}$$

$$\vec{u}_2: \vec{u}_2 = \langle u_1, x \rangle u_1 = \int_1^5 0,5 x \, dx \cdot 0,5 = 3$$

$$\|x-3\|^2 = \int_1^5 (x-3)^2 \, dx = \frac{16}{3}$$

$$\vec{u}_2 = \frac{x-3}{\sqrt{\frac{16}{3}}} = \frac{\sqrt{3}}{4}(x-3)$$

$$\begin{aligned} \vec{u}_3: \vec{u}_3 &= \langle \vec{u}_1, x^2 \rangle \vec{u}_1 + \langle \vec{u}_2, x^2 \rangle \vec{u}_2 \\ &= \int_1^5 0,5 x^2 \, dx \vec{u}_1 + \int_1^5 x^2 \cdot \frac{\sqrt{3}}{4}(x-3) \, dx \vec{u}_2 \\ &\quad x^2 - 6x + \frac{23}{3} \end{aligned}$$

$$\|x^2 - 6x + \frac{23}{3}\|^2 = \frac{256}{45}$$

$$\vec{u}_3 = \frac{\sqrt{45}}{16}(3x^2 - 18x + 23) \quad \left(\frac{1}{2}, \frac{\sqrt{3}}{4}(x-3), \frac{\sqrt{45}}{16}(3x^2 - 18x + 23) \right)$$

$$b) c = \langle \sin x, \vec{u}_1 \rangle = \int_1^5 \frac{1}{2} \sin x \, dx = \frac{1}{2} (\cos 1 - \cos 5)$$

$$d = \langle \sin x, \vec{u}_2 \rangle = \int_1^5 \frac{\sqrt{3}}{4} (x-3) \sin x \, dx = \frac{\sqrt{3}}{4} (\sin 5 - \sin 1 - 2 \cos 1 + 2 \cos 5)$$

$$a = \langle \sin x, \vec{u}_3 \rangle = \int_1^5 \frac{\sqrt{45}}{16} (3x^2 - 18x + 23) \sin x \, dx = \frac{\sqrt{45}}{8} (\cos 1 - \cos 5 + 6 \sin 1 + 6 \sin 5)$$

$$w(x) = c \vec{u}_1 + d \vec{u}_2 + a \vec{u}_3$$

$$w(x) = \frac{\cos 1 - \cos 5}{4} + \frac{\sqrt{3}(x-3)}{16} (\sin 5 - \sin 1 - 2 \cos 1 + 2 \cos 5) + \frac{5(3x^2 - 18x + 23)}{128} (\cos 1 - \cos 5 + 6 \sin 1 + 6 \sin 5)$$

$$c) \text{ min norm } \|\sin x - \underbrace{(ax^2 + bx + c)}_w\|$$

a) alle Pythagoren'sche: $\|\sin x - w\|^2 = \|\sin x\|^2 - \|w\|^2$ da $\sin x$ orthogonal zu w

$$\|\sin x - w\| \approx 0,235$$

